

测试与传感器技术（双语）

2023-2024 学年第二学期期中考试试题

共 100 分，50 分钟

- (1) List several commonly used sensors. (10 points)
- (2) The power dissipated in a car headlight is calculated by measuring the DC voltage drop across it and the current flowing through it. If the possible errors in the measured voltage and current values are $\pm 1\%$ and $\pm 2\%$ respectively, calculate the likely maximum possible error in the power value deduced. (20 points)
- (3) Find the autocorrelation function of $x(t) = e^{-at}$ ($a > 0, t \geq 0$). (20 points)
- (4) Find the mean square value ψ_x^2 , root mean square value x_{rms} , and auto-spectrum $S_x(\omega)$ of a signal which has autocorrelation function as $R_x(\tau) = 20\cos 20\pi\tau$. (25 points)
- (5) What is the steady state output $y(t)$ of a system with the transfer function $H(s) = \frac{1}{0.01s+1}$ and subjected to an input $x(t) = 0.6 \sin(10t) + 0.6 \sin(100t - 30^\circ)$? (25 points)